

Ten Layers of Gauge and Recurrence

Closed Algebra, Finite Barriers, and the Refined Physical Frontier

Prime Dynamics Theory Program

July 2026

Abstract

We audit RH-120–RH-129 as one proof system. Five layers are constructive, two establish obstructions, and three synthesize the route. Gauge-covariant Rayleigh transfer, optimal exact-Gram alignment, additive defect recurrence, normalization exponents, combined directional support, outward validation, and conditional eventual support are now exact. Two shortcuts are sharply blocked: forced common coordinates and unguarded independent assembly. Finite evidence separates the remaining mechanisms: all 24 regularized directional chains survive above 10^{-8} , whereas all 24 scalar direct margin chains lose positivity. Neither observation is asymptotic. The inclusion-minimal mathematical frontier is

$$\{P_D\}, \quad \{P_T\}, \quad \{P_R, P_A\},$$

where P_D is a physical direct recurrence, P_T a physical trace–concentration packet, P_R an all-level subunit Rayleigh recurrence, and P_A a positive normalized-base liminf. Independently assembled validation also requires all-level outward radii. None of these physical packets is proved.

1 Layer ledger

All nine upstream summaries pass machine-checked curated claims. In particular, RH-121 has 96 optimal-gauge pairs and 25 contractive tail factors; RH-125 has 24 regularized chains above 10^{-8} ; RH-126 has zero positive terminal direct chains; and RH-127 validates 4,096 random outward enclosures with zero false certifications.

2 What is now exact

The directional chain can be written without hidden algebra. RH-123 gives

$$x_{n+1} = \gamma_{n+1}^2 \leq \rho_n x_n + q_n.$$

RH-125 transports the determinant, leading scale, and capacity on the same gauge. RH-127 specifies the exact spectral-radius subtraction needed when the matrices come from

layer	class	principal result	route effect
120	constructive	gauge gamma/volume transfer	cross-level algebra opened
121	constructive	optimal exact-Gram gauge	intrinsic tail inflation exposed
122	negative	fixed-coordinate blow-up	no-gauge shortcut closed
123	constructive	affine defect recurrence	outward errors admitted
124	constructive	normalization exponents	separate-factor loss quantified
125	synthesis	combined directional transfer	regularized finite chains joined
126	negative	direct scalar-profile barrier	simple fallback held
127	constructive	outward Loewner guards	independent validation logic closed
128	synthesis	conditional liminf floor	directional algebra conditionally closed
129	synthesis	proof-frontier review	roadmap revised

Table 1: Primary classification of the ten layers.

independent assemblies. RH-128 then proves that eventual caps $\rho_n \leq \rho < 1$, $q_n \leq q$, with $q/(1 - \rho) < 1$, and a positive liminf for $A_n = V_n/(L_n^4 C_n)$ imply

$$\liminf B_n \geq \left(1 - \sqrt{\frac{q}{1 - \rho}}\right)^4 \liminf A_n > 0.$$

These steps use standard positive-matrix, singular-value, and interval arguments [Horn and Johnson, 1991, Moore, 1966]. The implication is exact; its physical premises are open.

3 Finite barriers and non-implications

RH-122 proves a genuine universal obstruction: identical generalized tail spectra can have fixed-coordinate transfer loss $1/\varepsilon$. Thus a gauge or common-coordinate coherence law is mandatory. RH-127 closes a second proof-method gap: floating-point agreement across independent paths is not an inequality, while outward radii provide one.

The RH-125 and RH-126 comparisons are finite diagnostics, not universal theorems about future scales. The regularized directional chains are encouraging but use an explicit positive Gram conditioning floor. The direct scalar-profile chains fail, but a richer operator-valued recurrence is not excluded. RH-117 already forbids turning either finite pattern into an all-level law by interpolation.

4 Refined proof frontier

Let E denote eventual support. Relative to the audited rules, its current physical production rules are

$$P_D \Rightarrow E, \quad P_T \Rightarrow E, \quad (P_R \wedge P_A) \Rightarrow E.$$

For an independently assembled validated conclusion E_{val} , each rule additionally requires P_O , the all-level outward-radius packet.

Theorem 1 (Refined frontier antichain). *From the currently proved nodes, E is unreachable. Its inclusion-minimal missing sets are exactly*

$$\{P_D\}, \quad \{P_T\}, \quad \{P_R, P_A\}.$$

For E_{val} they are

$$\{P_D, P_O\}, \quad \{P_T, P_O\}, \quad \{P_R, P_A, P_O\}.$$

Proof. None of the physical nodes is proved. Supplying each displayed set completes one rule, while no proper subset completes any rule. Exhaustive closure over all subsets of the five physical nodes returns exactly these antichains. $\square$

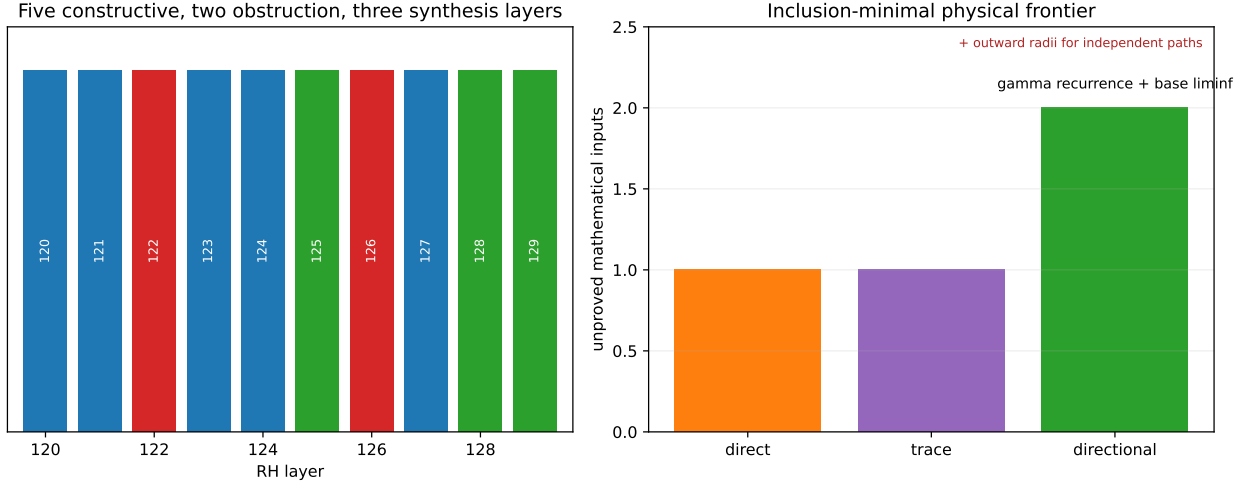


Figure 1: The ten-layer classification and the refined mathematical frontier.

5 Revised route

The next finite gate should rebuild RH-121/RH-125 on one high-precision or interval assembly, remove the positive Gram floor, work on the actual supported subspace, and archive RH-127 radii. If the directional advantage survives, the theoretical priority is a natural gauge derived from the memory and packet recursions, followed by source-level bounds on ρ_n , q_n , and the normalized-base liminf. The scalar direct and trace routes remain on hold pending genuinely new source laws.

No all-level physical packet, uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.
- Ramon E. Moore. *Interval Analysis*. Prentice-Hall, 1966.